Consumer Behavior

Chapters 4

1 The Consumer's Preferences and the Concept of Utility

1.1 Assumptions about Consumer Preferences

> **Consumption Bundle:** A set of goods or services a consumer considers purchasing.

1. Completeness and Rankability : Consumers can compare bundles of goods and rank them based on preference.
2. For most goods, more is better than less: Non-satiation and free disposal
3. Transitivity : Imposes logical consistency on preferences
   - A is preferred to B
   - B is preferred to C
   - Then, A must be preferred to C
4. Love of variety : The more a consumer has of a particular good, the less she is willing to give up of something else to get even more of that good.

1.2 The Concept of Utility

> **Utility:** A measure of how satisfied a consumer is.

- A measure of happiness or well-being .
- It is not a measure of consumer income .

> **Utility Function:** A mathematical function that describes the relationship between what consumers actually consume and their level of well-being.

- Represents consumer preferences .
- Can take a variety of mathematical forms.
- Have to conform to the four assumptions about preferences.

$$U = U(J, K)$$

Examples: Suppose we have two goods: Junior Mints (J) and Kitkat bars (K)

- $U = J/K$
  - Violates more is better
- $U = J + K$
  - Violates love of variety
  - Do an example showing this
- $U = K^{\alpha} J^{\beta}$
  - Conforms to the assumptions
- $U = KJ + K + J$
  - Conforms to the assumptions

1.3 Utility and Comparisons

The "rules" for utility allows only for an ordinal ranking of consumption bundles.

- An ordinal ranking implies bundles can be ranked from best to worse.
- A cardinal ranking would allow a person to determine how much better one bundle is, compared to another.

Why not cardinal?

- Many questions can be answered with only an ordinal ranking (e.g. predicting what will be consumed)
- There is no real-world measure of how much more a consumer likes the bundle of goods A to the bundle of goods B.

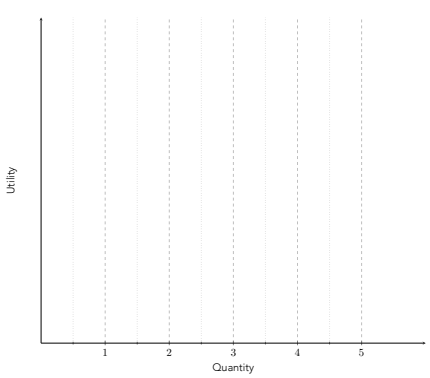
The thing we can't do, though, is make interpersonal comparisons of utility, meaning one person's utility to some other person's.

1.4 Marginal Utility

> **Marginal Utility:** The extra utility a consumer receives from a 1-unit increase in consumption.

$$MU_K = \frac{\Delta U(J,K)}{\Delta K} = \frac{\partial U(J,K)}{\partial K}$$

Draw a graph showing the diminishing total utility curve. Then show diminishing marginal utility.



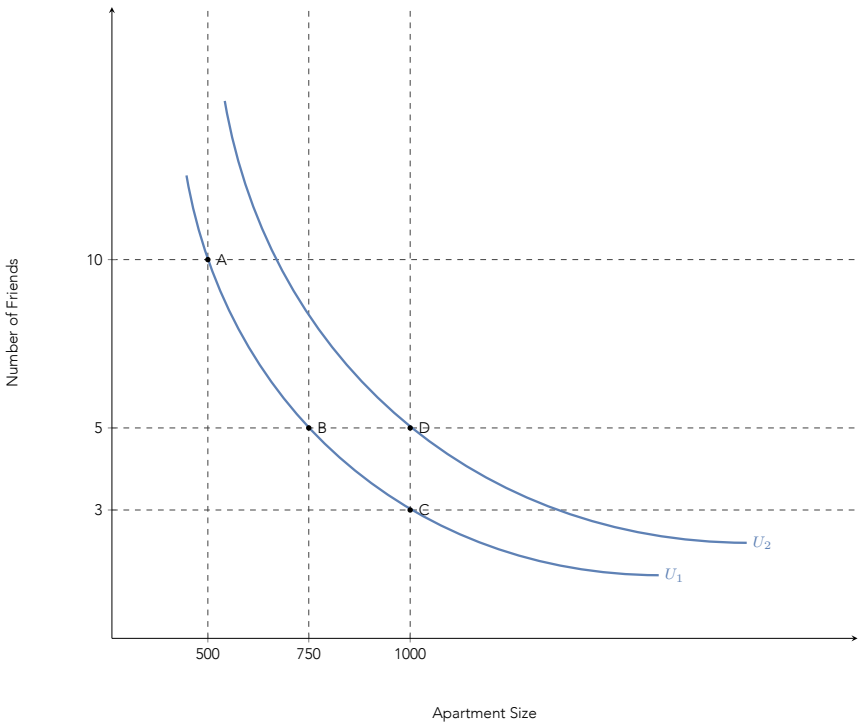
POLL EVERYWHERE: The additional utility obtained from consuming an extra unit of a good is called:

2 Indifference Curves

> **Indifferent:** The special case in which a consumer derives the same level of utility from each of two or more consumption bundles.

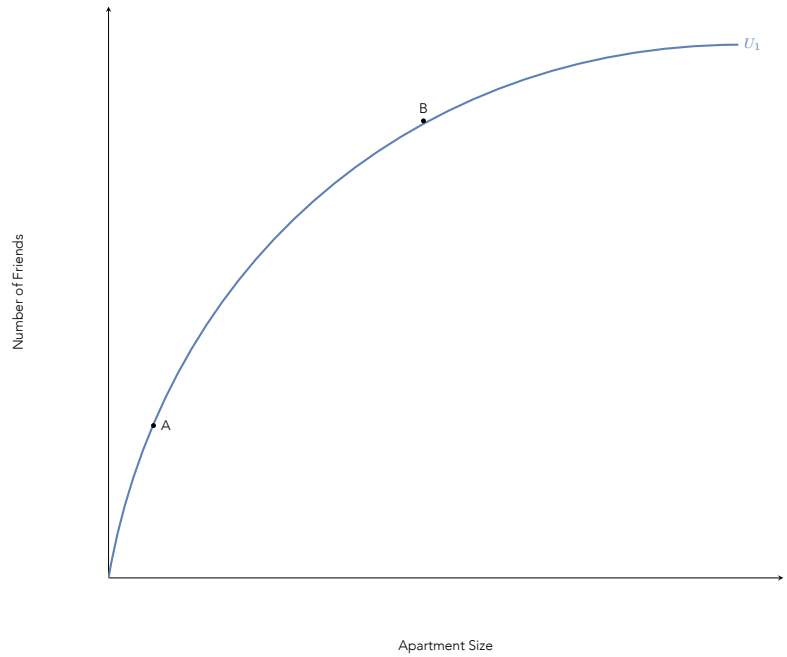
> **Indifference Curve:** The combination of all the different consumption bundles that provide a consumer with the same utility.

Consider the case of renting an apartment where there are two "goods" you must choose between: the size in square feet and the number of friends living in the same building. Michaela wants a large apartment, but also wants to be able to easily see her friends.

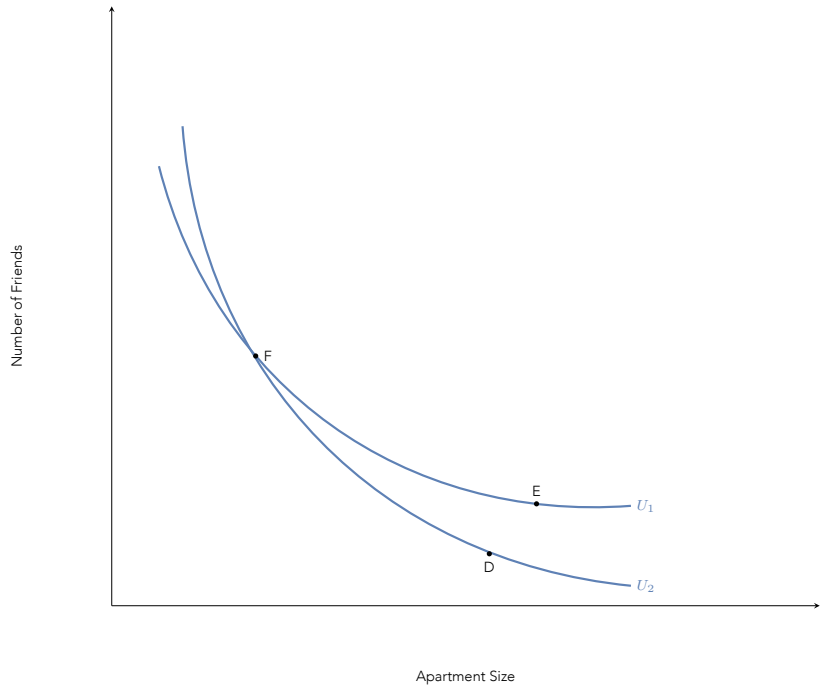


2.1 Characteristics of Indifference Curves

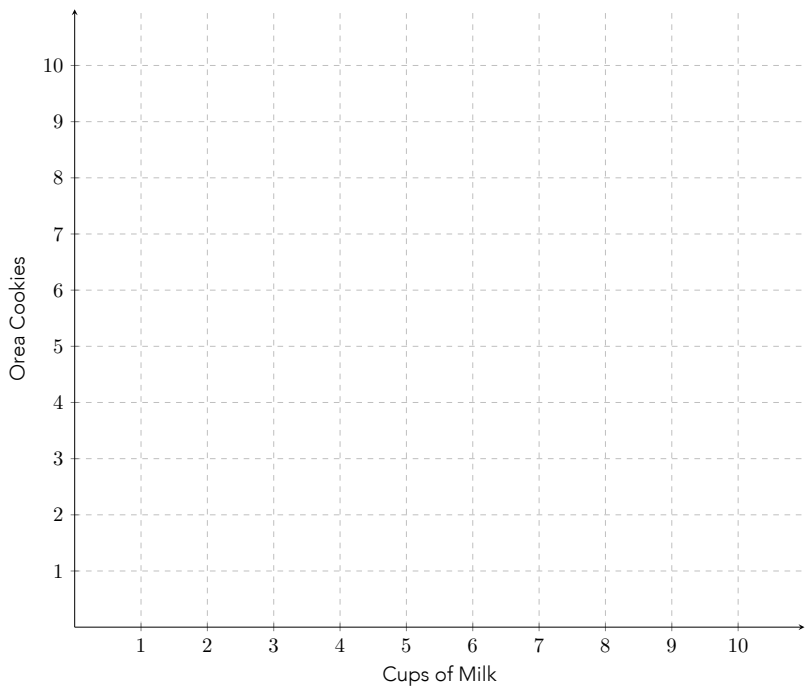
1. We can always draw indifference curves. Rankability and Completeness .
2. We can figure out which indifference curves have higher utility levels
   - More Utility: More is better
   - As indifference curves move away from the origin, the utility they represent is increasing
3. We know why they slope downward .
   - Suppose we have an upward sloping indifference curve



   - Because A and B are on the same indifference curve, they give us the same level of utility .
   - But B has more of both goods than A , so we should prefer B to A .
   - Therefore an upward sloping indifference curve does not make sense.
4. Indifference curves never cross .
   - Because F and D are on indifference curve U_1 they give us the same utility (we are indifferent).
   - Because F and E are on indifference curve U_2 they give us the same utility (we are indifferent).
   - Transitivity would imply that since we are indifferent between F and D and we are indifferent between F and E , we should be indifferent between D and E .
   - But E has more of both goods than D, so we would prefer E to D (more is better).
   - Therefore, crossing indifference curves violates transitivity.



5. Indifference curves are convex to the origin (i.e., they bend toward the origin in the middle and away at the edges).
   - Love of variety



POLL EVERYWHERE: Which assumption(s) allow(s) us to draw indifference curves?

2.2 The Marginal Rate of Substitution

How much of one good will a consumer give up to get a little bit more of another good?

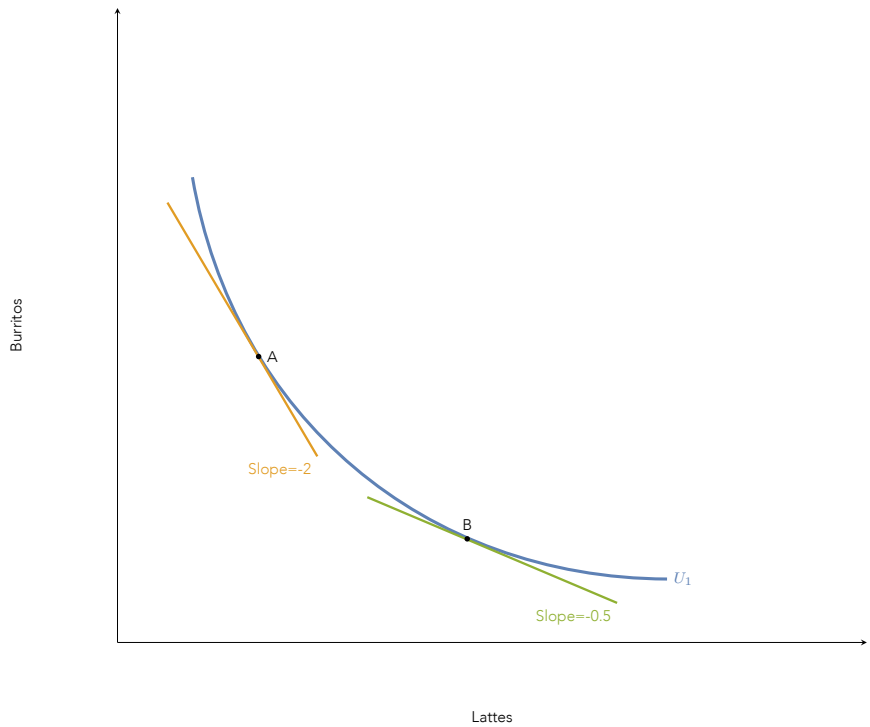
- The slope of the indifference curve exactly captures this tradeoff idea.

> **Marginal Rate of Substitution:** The rate at which a consumer is willing to trade off one good (the good on the horizontal axis, X) for another (the good on the vertical axis, Y) and still be left

> equally well off.

$$MRS_{XY} = -\frac{\Delta Y}{\Delta X}$$

Consider Sarah's indifference curve:



2.3 The Marginal Rate of Substitution and Marginal Utility

Let's still consider Sarah's burritos and lattes.

- At point A: $MRS_{BL} = -\frac{\Delta Q_B}{\Delta Q_L} = -2$, i.e. 2 burritos for 1 more latte.
- At point B: $MRS_{BL} = -\frac{\Delta Q_B}{\Delta Q_L} = -0.5$, i.e. 0.5 burritos for 1 more latte.

This change in the willingness to substitute between goods at the margin occurs because the benefit a consumer gets from another unit of a good falls with the number of units she already has. If you have already amped up on caffeine after drinking 6 lattes but haven't eaten all day, you'd forgo lots of lattes to get a burrito.

Another way to see all of this is to think about the change in utility (ΔU) created by starting at some point on an indifference curve and moving a short distance along it.

$$\Delta U = MU_L \times \Delta Q_L + MU_B \times \Delta Q_B$$

When we move along the indifference curve, we know the at the overall level of utility is not changing ($\Delta U = 0$).

$$0 = MU_L \times \Delta Q_L + MU_B \times \Delta Q_B$$

Rearranging yields:

$$\underbrace{-\frac{\Delta Q_B}{\Delta Q_L}}_{MRS_{BL}} = \frac{MU_L}{MU_B}$$

The MRS_{XY} between two goods at any point on an indifference curve equals the inverse ratio of those two goods' marginal utilities

You can tell how much people value something by their choices of what they would be willing to give up to get it. The rate at which they give things up reveals the marginal utility of the goods.

2.4 The Steepness of Indifference Curves

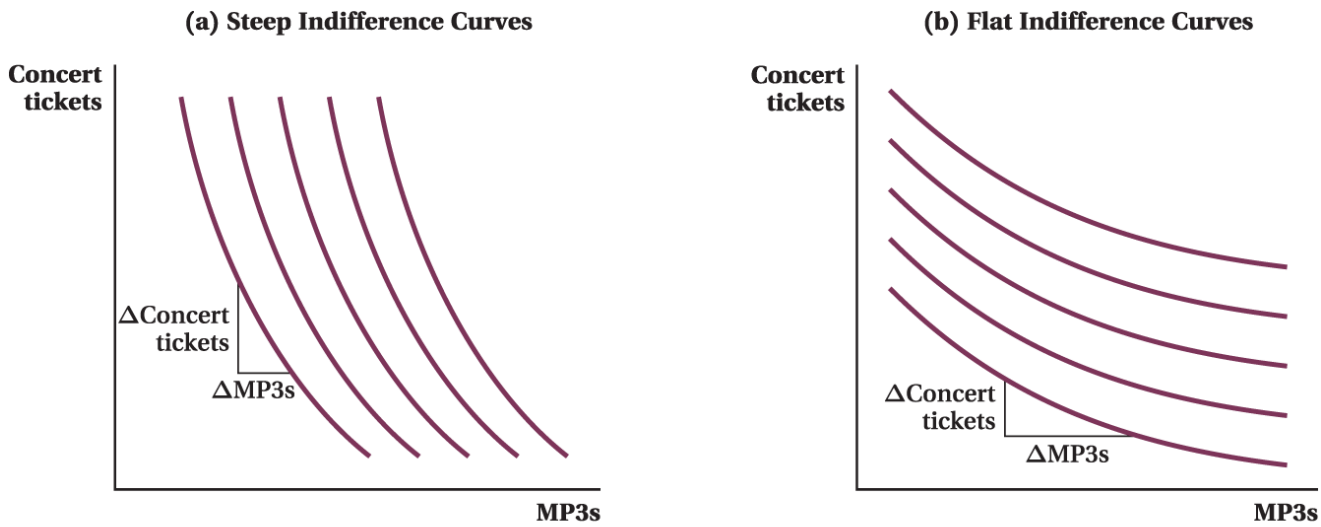


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Figure 1

Because the MRS measures the willingness of the consumer to trade one good for another, we can tell a great deal about preferences by examining the shapes of indifference curves. (a) Indifference curves that are relatively steep indicate the consumer is willing to give up a large quantity of the good on the vertical axis to get another unit of the good on the horizontal axis. Here, the consumer is willing to give up a lot of concert tickets for some additional MP3s. (b) Relatively flat indifference curves imply that the consumer would require a large increase in the good on the horizontal axis to give up a unit of the good on the vertical axis. The consumer with flat indifference curves will give up a lot of MP3s for one additional concert ticket.

2.5 The Curvature of Indifference Curves: Substitutes and Complements

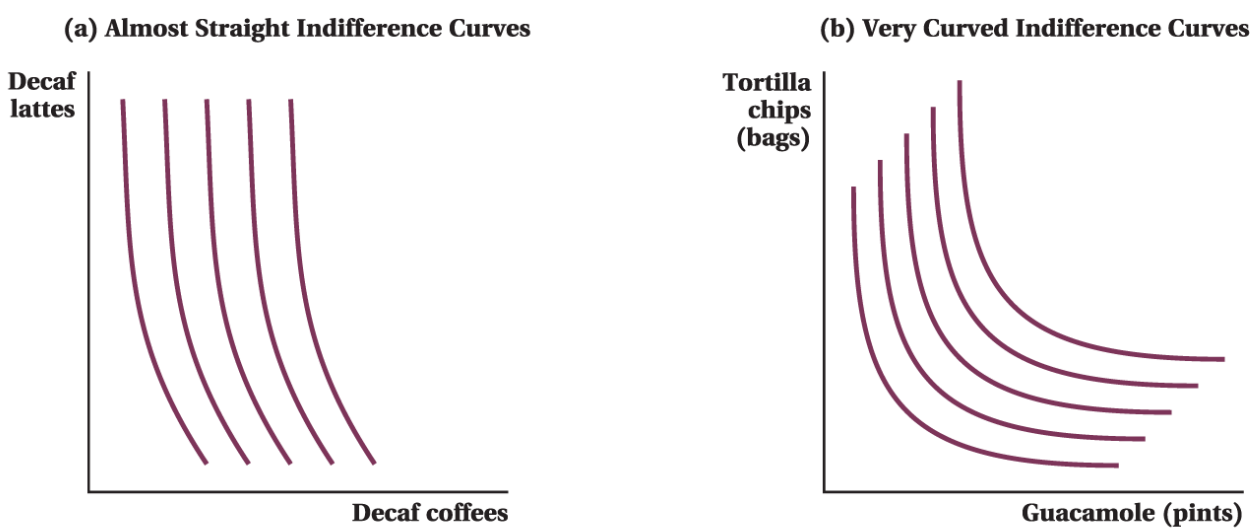

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Figure 2

The curvature of indifference curves reflects information about the consumer's preferences between two goods, just as its steepness does. (a) Goods that are highly substitutable (such as decaf lattes and decaf coffees) are likely to produce indifference curves that are relatively straight. This means the MRS does not change much as

the consumer moves from one point to another along the indifference curve. (b) Goods that are complementary will generally have indifference curves with more curvature. For example, if a consumer has many tortilla chips and little guacamole, he will be willing to trade many tortilla chips to get some guacamole. If a consumer has a lot of guacamole and few tortilla chips, he will be less willing to trade chips for guacamole.

2.5.1 Perfect Substitutes

> **Perfect Substitutes:** A good that a consumer can trade for another good, in fixed units, and receive the same level of utility.

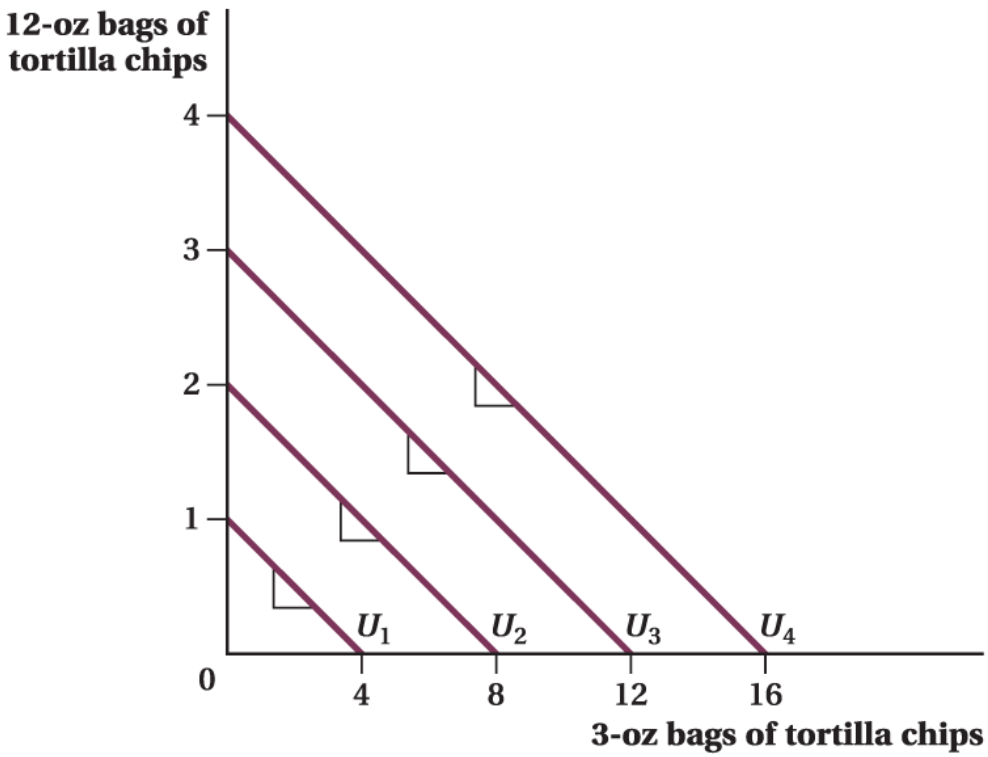

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Figure 3

For perfect substitutes, the indifference curves are straight lines with a constant slope, in this case equal to -1/4, which means that the MRS_{XY} is also constant and equal to 1/4. We can't actually say what values a and b take here, only that their ratio is 1 to 4 — that is, a/b = 1/4. The indifference curves in this example would be the same if a = 1 and b = 4 or if a = 40 and b = 160, for instance. This is another demonstration of the point we made above: A transformation of a utility function that does not change the order of which goods the consumer prefers implies the same preference choices.

2.5.2 Perfect Complements

> **Perfect Complements:** A good from which the consumer receives utility dependent on its being used in a fixed proportion with another good.

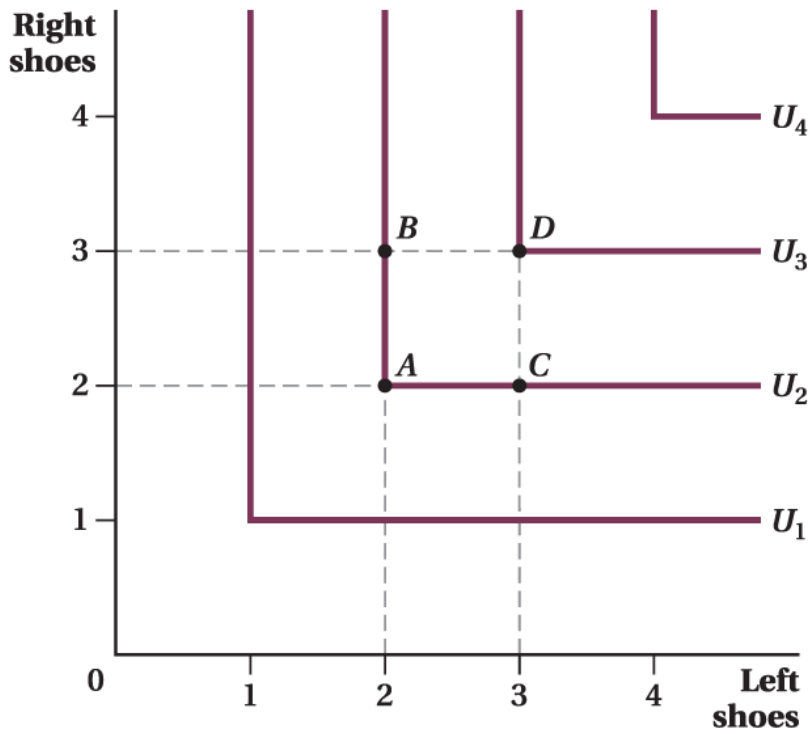

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Figure 4

When goods are perfect complements, they have L-shaped indifference curves. For example, at point A, the consumer has 2 left shoes and 2 right shoes. Adding another right shoe while keeping the left shoes constant does not increase the consumer's utility, so point B is on the same indifference curve as point A. In like manner, adding another left shoe will not increase the consumer's utility without an additional right shoe, so point C is on the same indifference curve as points A and B. Because shoes are always consumed together, 1 right shoe and 1 left shoe, the consumer's utility rises only when she has more of both goods (a move from point A to point D).

Mathematically this can be represented as: $U = min(aX, bY)$, where a and b are again numbers reflecting how consuming more units of X and Y affects utility.

2.5.3 Different Shapes for a Particular Customer

Different Consumers may have differently shaped indifference curves.

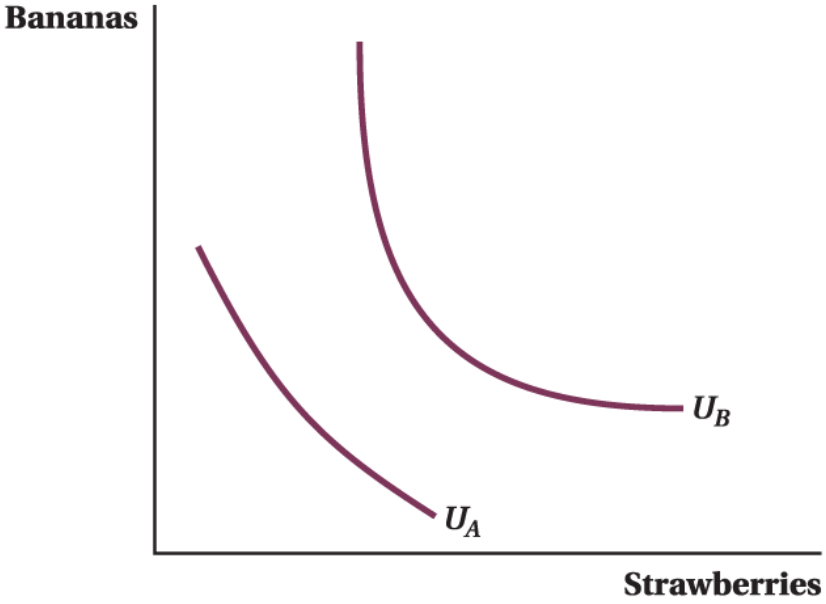

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Figure 5

3 The Consumer's Income and the Budget Constraint

Our ultimate goal in this chapter is to understand how consumers maximize their utility by choosing a bundle of goods to consume. Because consumers have a limited amount of money and because goods are not free, consumers must make tradeoffs when deciding how much of each good to consume.

Start with three assumptions:

1. Each good has a fixed price, and any consumer can buy as much of a good as she wants at that price if she has the income to pay for it. We can make this assumption because each consumer is only a small part of the market for a good, so her consumption decision will not affect the equilibrium market price.
2. The consumer has a fixed amount of income available to spend.
3. For now, the consumer cannot borrow or save. Without borrowing, she can't spend more than her income in any given period. With no saving, it means that unspent money is lost forever, so it's use it or lose it.

> **Budget Constraint:** A curve that describes the entire set of consumption bundles a consumer can purchase when spending all income.

Suppose Sarah has $50 to spend on burritos (which cost $10 each) and lattes ($5 each).

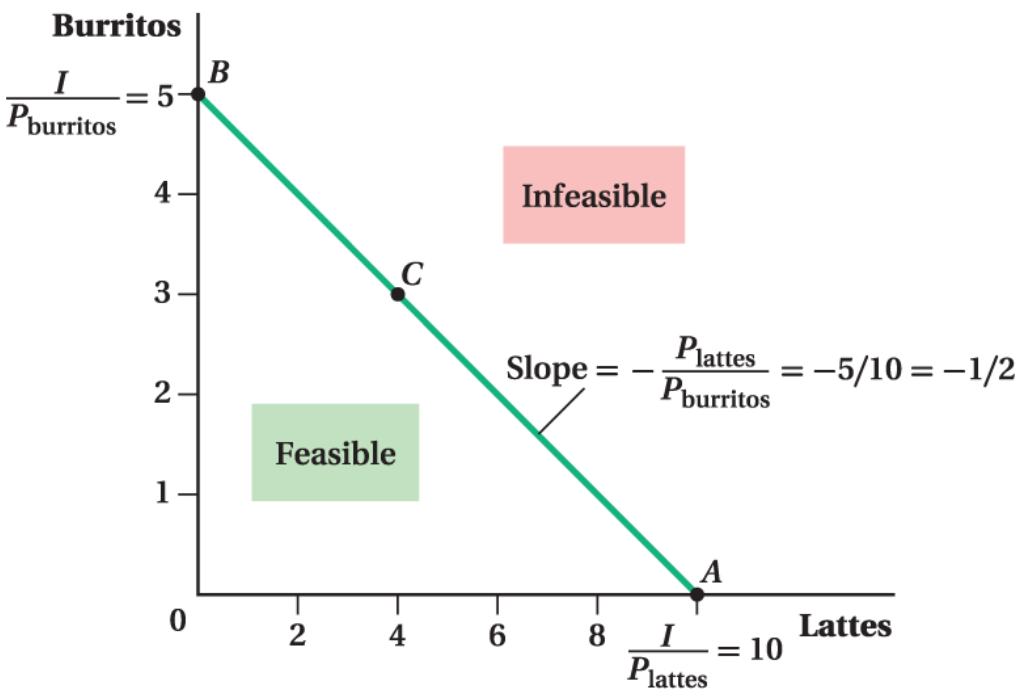

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Figure 6

The mathematical formula for a budget constraint is:

$$Income = P_X Q_X + P_Y Q_Y$$

> **Feasible Bundle:** Any combination of goods on or below the budget constraint that the consumer has the income to purchase.

> **Infeasible Bundle:** Any combination of goods above or to the right of the budget line that the consumer cannot afford to purchase.

POLL EVERYWHERE: Beth has $400 of income to spend on e-books (E), priced at $20 per book, and exercise classes called Zumba (Z), priced at $16 per class. Which of the following consumption bundles is infeasible?

3.1 The Slope of the Budget Constraint

$$Income = P_X Q_X + P_Y Q_Y$$
$$Q_Y = \frac{Income}{P_Y} - \frac{P_X}{P_Y} Q_X$$

If X is expensive relative to Y (P_X/P_Y is large), then you must give up a lot of Y to get more X.

Consider Sarah:

$$50 = 5Q_L + 10Q_B$$
$$Q_B = 5 - \frac{1}{2}Q_L$$

3.2 Factors That Affect the Budget Constraint

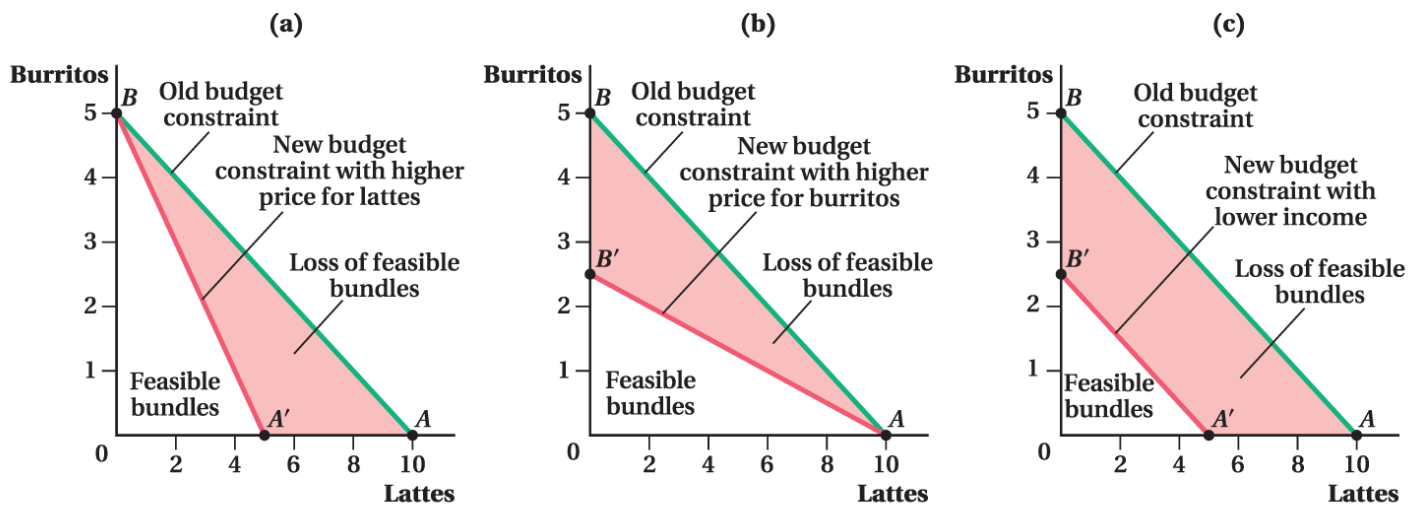

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Figure 7

(a) Price of lattes increases.

(b) Price of burritos increases.

(c) Income falls.

3.3 Nonstandard Budget Constraints

3.3.1 Quantity Discounts

Suppose Alex spends his $100 weekly income on pizzas and data usage on his phone. A pizza costs $10; if he spends everything on pizzas, he can buy 10. If the price of phone data is constant at 10 cents per MB, he can buy up to 1 GB (1,000 MBs) of data.

Phone plans often offer quantity discounts on data. With a quantity discount, the price the consumer pays per unit of the good depends on the number of units purchased. If Alex's calling plan charges 10 cents per MB for the first 600 MBs per month and 5 cents per MB after that, his budget constraint will have a kink.

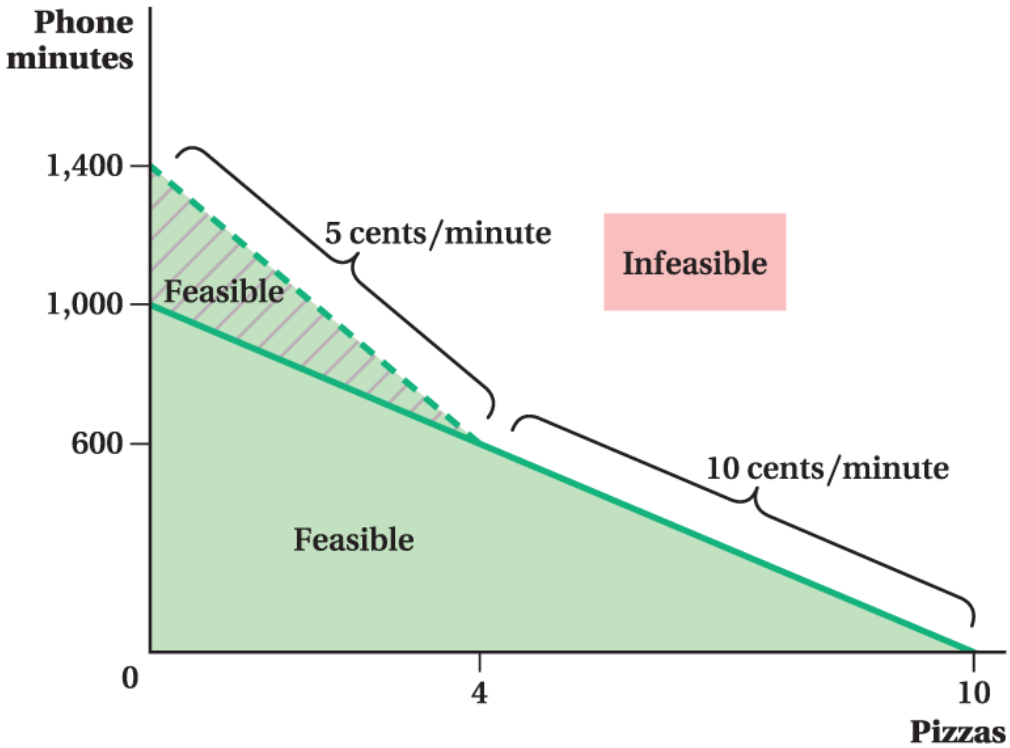

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Figure 8

3.3.2 Quantity Limits

Suppose that Alex's carrier or the government (or maybe his parents) expressly capped Alex's weekly data usage at 600 MBs per week. In that case, the part of the budget constraint beyond 600 MB becomes infeasible, and the constraint becomes horizontal at 600, as shown by the solid line in Figure 4.17.

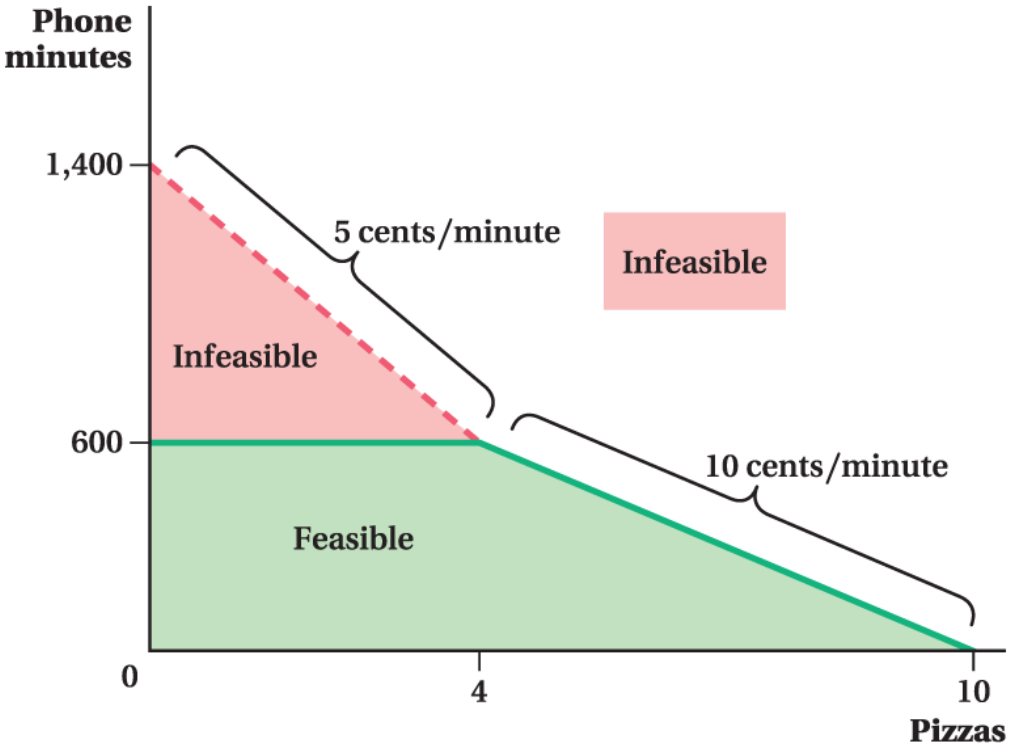

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Figure 9

4 Combining Utility, Income, and Prices, What Will the Consumer Consume?

4.1 Solving the Consumer's Optimization Problem

Before we try to solve this constrained optimization problem, think about what makes it tricky: We must compare some things measured in dollars (like income and prices) to other things measured in imaginary units that don't translate into dollars (like utils of utility). How can you know whether you're willing to pay $3 to get an extra unit of utility? Well, you can't, really. What you can figure out, however, is whether spending an extra dollar on, say, golf balls gives you more or less utility than spending an extra dollar on something else, like AAA batteries. It turns out that figuring out this choice using indifference curves and budget constraints makes solving the consumer's optimization problem straightforward.

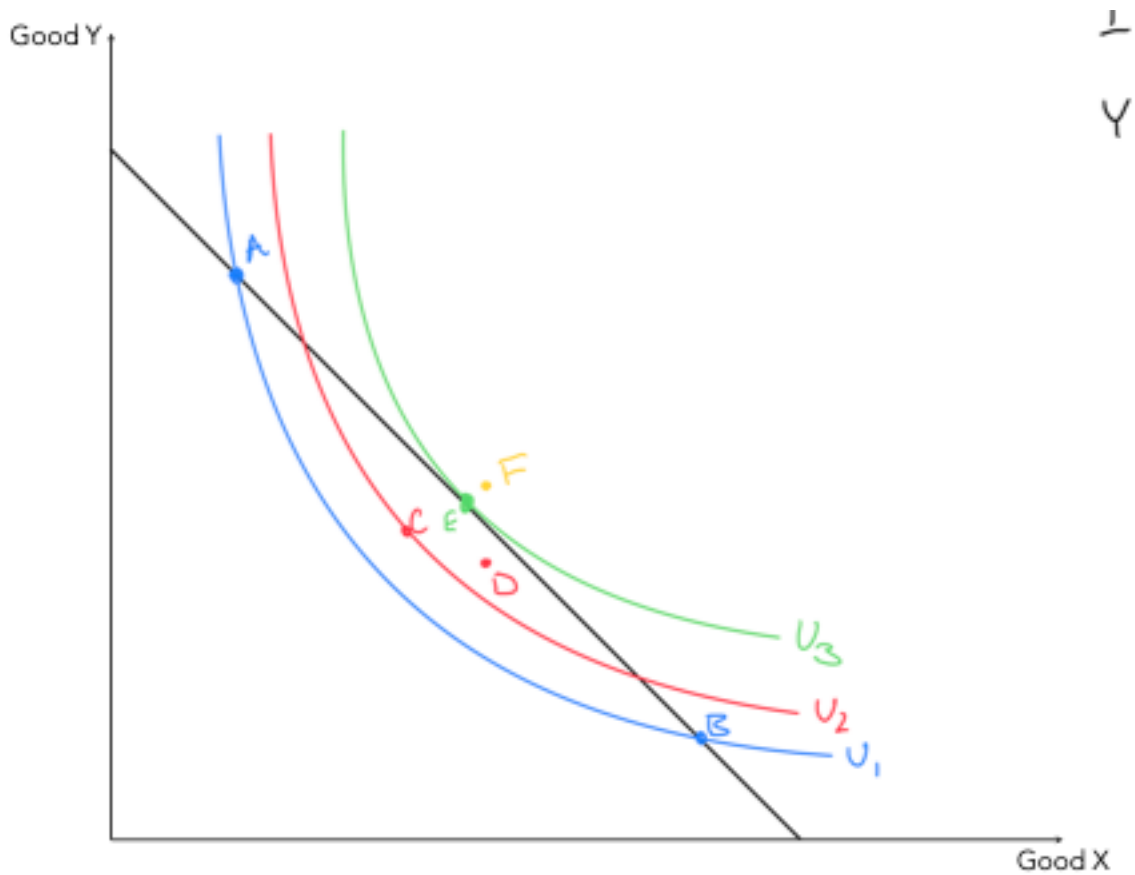

Figure 10

- A and B are efficient, but not optimal because C is feasible and lies on a higher indifference curve (C is preferred to A and B).
- C is feasible, but inefficient, therefore there is always another consumption bundle (for instance: D) that will be preferred to C.
  - Therefore the optimal consumption bundles must be efficient.
- E is both feasible and efficient. It is also the optimal consumption bundles. Why?
  - Out of all the feasible and efficient consumption bundles, E is on the highest indifference curve.
  - All consumption bundles that are preferred to E (for instance: F), not feasible.

When the consumer spends all her income and maximizes her utility, her optimal consumption bundle is the one at which the ratio of the goods' marginal utilities exactly equals the ratio of their prices.

$$\text{Slope of the indifference curve} = \text{Slope of the budget constraint}$$
$$-MRS_{XY} = -P_X/P_Y$$
$$MU_X/MU_Y = P_X/P_Y$$
$$MU_X/P_X = MU_Y/P_Y$$

Note, the utility maximizing point is when the marginal utilities per dollar are equal to each other. Here, the utility-maximization problem can be restated as finding the consumption bundle that gives the consumer the most bang for her buck.

POLL EVERYWHERE: Rohan is maximizing utility subject to his limited income. As a result, which of the following statements is TRUE?

- Suppose $\frac{MU_X}{P_X} > \frac{MU_Y}{P_Y}$:
  - You should buy more of good X and less of good Y, because you get more additional utility (as measured by marginal utility) per dollar from good X than you do from good Y.
  - This will cause MU_X to fall and MU_Y to rise (because of diminishing marginal utility).
  - Eventually the falling MU_X and rising MU_Y will cause $\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y}$.
- Suppose $\frac{MU_X}{P_X} < \frac{MU_Y}{P_Y}$:
  - You should buy more of good Y and less of good X, because you get more additional utility (as measured by marginal utility) per dollar from good Y than you do from good X.
  - This will cause MU_Y to fall and MU_X to rise (because of diminishing marginal utility).
  - Eventually the falling MU_Y and rising MU_X will cause $\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y}$.

4.2 Implications of Utility Maximization

Even if two consumers have different preferences over the same goods, if they face the same prices, they will have the same MRS at their optimal consumption bundle.

Even if two consumers have very different preferences between two goods, they will have the same ratio of marginal utilities for the two goods in equilibrium, because utility maximization implies that the MRS for each of them will equal the ratio of the prices, and that ratio is the same for both of them.

Consider Meg and Jack:

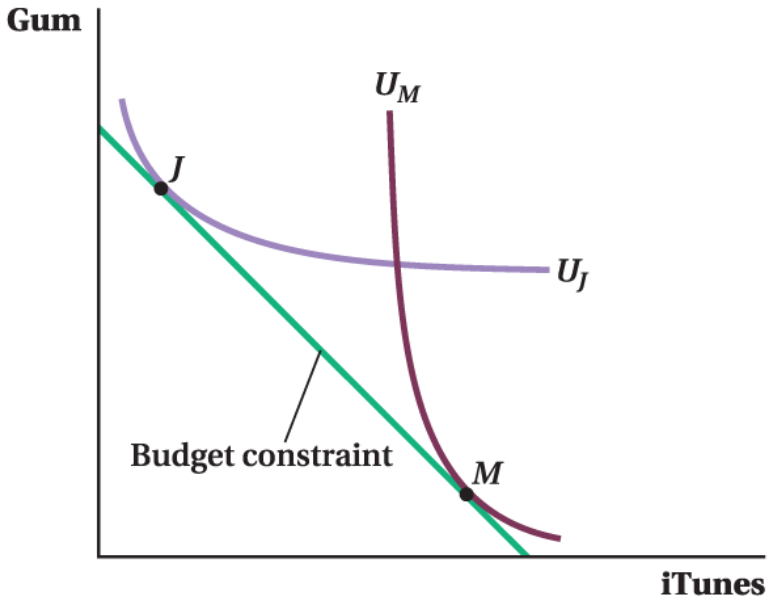

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Figure 11

4.3 A Special Case: Corner Solutions

> **Corner Solution:** A utility-maximizing bundle located at the "corner" of the budget constraint where the consumer purchases only one of two goods.

> **Interior Solution:** A utility-maximizing bundle that contains positive quantities of both goods.

Greg, our consumer, has an income of $240 and is choosing his consumption levels of mystery novels and Bluetooth speakers. Let's say a hardcover mystery novel costs $20, and a Bluetooth speaker costs $120. Because speakers are more expensive than mystery novels, Greg can afford up to 12 mysteries but only 2 speakers. Nonetheless, the highest utility that Greg can obtain given his income is bundle A, where he consumes all speakers and no mystery novels.

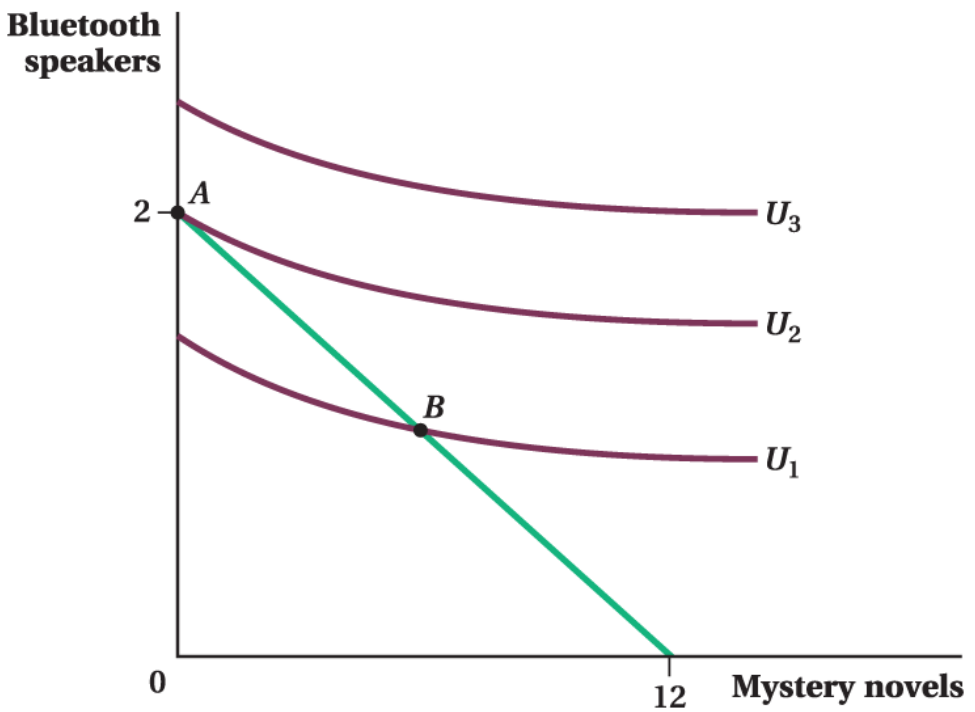

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Figure 12

In a corner solution, then, the highest indifference curve touches the budget constraint exactly once, just as with the interior solutions we discussed earlier. The only difference with a corner solution is that bundle A is not a point of tangency. The indifference curve is flatter than the budget constraint at that point (and everywhere else). That means Greg's MRS — the ratio of his marginal utility from mystery novels relative to his marginal utility from Bluetooth speakers — is less than the price ratio of the two goods rather than equal to it. In other words, even when he's consuming no mysteries, his marginal utility from them is so low, it's not worth paying the price of a book to be able to consume one. The marginal utility he'd have to give up with fewer speakers would not be made up for by the fact that he could spend some of his money on novels.

5 Appendix: The Calculus of Utility Maximization and Expenditure Minimization

5.1 Consumer's Optimization Problem

- The consumer maximizes their utility $U(X, Y)$ when:

$$MRS_{XY} = \frac{MU_X}{MU_Y} = \frac{P_X}{P_Y}$$

- We can use calculus to derive the marginal utilities from X and Y.

5.2 Utility Maximization

- The preference side (MRS_{XY}) is one side of the consumer's problem. We also need to know about the consumer's problem.
- Suppose the consumer's utility is:

$$U(X, Y) = X^{\alpha} Y^{1-\alpha}$$

- And the consumer's budget constraint is:

$$I = P_X X + P_Y Y$$

- Therefore the consumer's utility-maximization problem is:

$$\max_{X,Y} U(X, Y) = X^{\alpha} Y^{1-\alpha}$$
$$\text{s.t. } I = P_X X + P_Y Y$$

- This is known as a constrained optimization problem.
- How do we solve this problem?

5.2.1 Solving for the optimal point

- From above we know that for a consumer, the optimal point occurs when:

$$MRS_{XY} = \frac{P_X}{P_Y}$$

- Start by finding the marginal utilities of X and Y:

$$MU_X = \frac{\partial U(X, Y)}{\partial X} = \frac{\partial X^{\alpha} Y^{1-\alpha}}{\partial X} = \alpha X^{\alpha-1} Y^{1-\alpha}$$
$$MU_Y = \frac{\partial U(X, Y)}{\partial Y} = \frac{\partial X^{\alpha} Y^{1-\alpha}}{\partial Y} = (1-\alpha) X^{\alpha} Y^{-\alpha}$$

- Substitute these two marginal utilities into the formula for the marginal rate of substitution:

$$MRS_{XY} = \frac{MU_X}{MU_Y} = \frac{\alpha X^{\alpha-1} Y^{1-\alpha}}{(1-\alpha) X^{\alpha} Y^{-\alpha}} = \frac{\alpha}{1-\alpha} \frac{Y}{X}$$

- Set the marginal rate of substitution equal to the ratio of the prices:

$$MRS_{XY} = \frac{\alpha}{1-\alpha} \frac{Y}{X} = \frac{P_X}{P_Y}$$
$$Y = \frac{(1-\alpha) P_X}{\alpha P_Y} X$$

- Plug this relationship between Y and X into the budget constraint:

$$
\begin{aligned}
I &= P_X X + P_Y Y \\
I &= P_X X + P_Y \left[\frac{(1-\alpha)P_X}{\alpha P_Y} X \right] \\
I &= P_X X + \frac{1-\alpha}{\alpha} P_X X \\
I &= \frac{P_X}{\alpha} X \\
X &= \frac{\alpha I}{P_X} \\
Y &= \frac{(1-\alpha)P_X}{\alpha P_Y} \left(\frac{\alpha I}{P_X} \right) = \frac{(1-\alpha)I}{P_Y}
\end{aligned}
$$

- The optimal solution is dependent on all three pieces of the consumer's problem:
  - Consumer Preferences: α
  - Consumer Income: I
  - The good's prices: (P_X, P_Y)

5.2.2 Example #1

- Suppose we have the following problem:

$$
\begin{aligned}
&\max_{B,F} U(B,F) = B^{0.5} F^{0.5} \\
&\text{s.t. } \$5B + \$2F = \$20
\end{aligned}
$$

- Start by finding the marginal utilities for burgers and fries.

$$
\begin{aligned}
MU_B &= 0.5 B^{-0.5} F^{0.5} \\
MU_F &= 0.5 B^{0.5} F^{-0.5}
\end{aligned}
$$

- Find the marginal rate of substitution:

$$
MRS_{BF} = \frac{MU_B}{MU_F} = \frac{F}{B}
$$

- Set the MRS equal to the ratio of the prices:

$$
\begin{aligned}
\frac{F}{B} &= \frac{5}{2} \\
F &= 2.5B
\end{aligned}
$$

- Plug it into the budget constraint:

$$
\begin{aligned}
20 &= 5B + 2F \\
20 &= 5B + 2(2.5B) \\
20 &= 10B \\
B &= 2 \\
F &= 2.5 \times 2 = 5
\end{aligned}
$$

5.2.3 Example #2

- Suppose we have the following problem:

$$\max_{B,F} U(B,F) = BF + 2B + 4F$$
$$\text{s.t. } \$5B + \$2F = \$20$$

- Start by finding the marginal utilities for burgers and fries.

$$MU_B = F + 2$$
$$MU_F = B + 4$$

- Find the marginal rate of substitution:

$$MRS_{BF} = \frac{MU_B}{MU_F} = \frac{F+2}{B+4}$$

- Set the MRS equal to the ratio of the prices:

$$\frac{F+2}{B+4} = \frac{5}{2}$$
$$2F + 4 = 5B + 20$$
$$2F = 5B + 16$$
$$F = 2.5B + 8$$

- Plug it into the budget constraint:

$$20 = 5B + 2F$$
$$20 = 5B + 2(2.5B + 8)$$
$$20 = 10B + 16$$
$$4 = 10B$$
$$B = 0.4$$
$$F = 2.5 \times 0.4 + 8 = 9$$